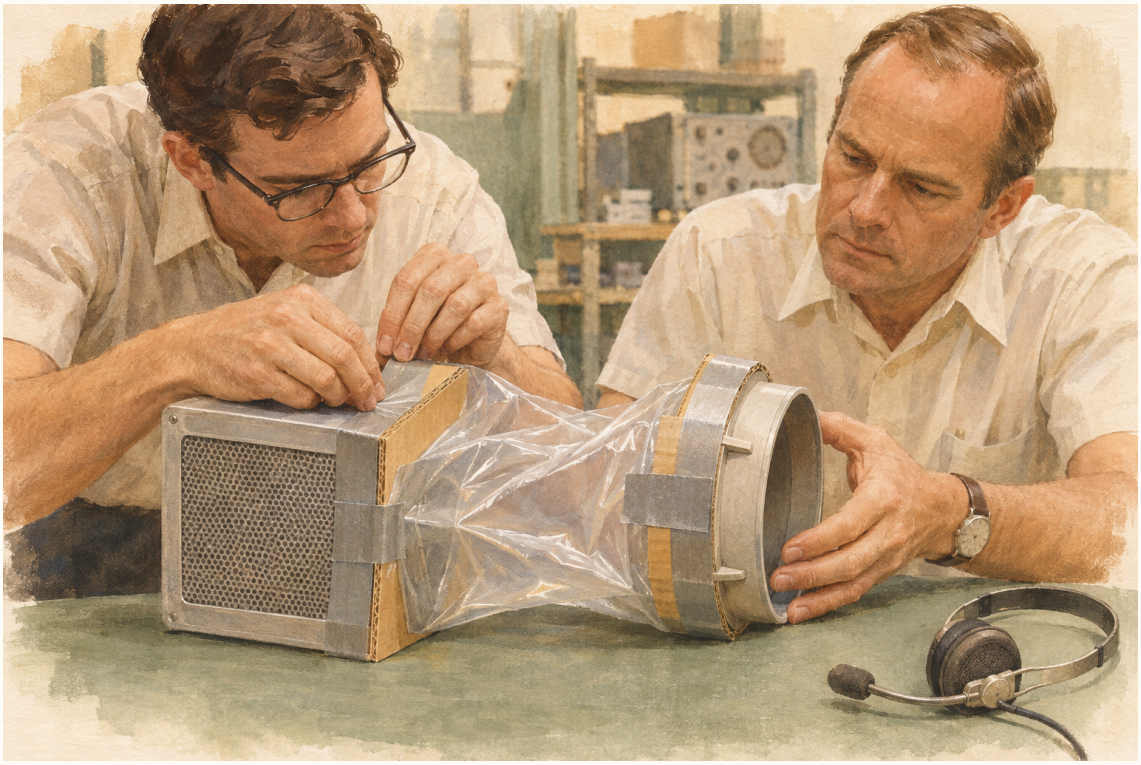
阿波罗13号：成功的失败

Apollo 13: The Successful Failure



方形滤芯怎样接进圆形插槽，救回三名宇航员？

英文原文

On April 11th 1970 NASA launched Apollo 13, meant for the third crewed lunar landing. On April 13th, some 200,000 miles out, an oxygen tank exploded in the service module, home to most of the ship's oxygen and power. The blast crippled the spacecraft; the landing was aborted.

The astronauts took refuge in the lunar module, Aquarius, built to support two men for about two days. Using its engine, the crew steered the craft onto a trajectory around the Moon, whose gravity swung it home. They rationed water, conserved power, and shivered in a bitterly cold cabin.

A quieter hazard emerged: with a third man aboard, carbon dioxide was accumulating. The command module, the crew's capsule, used square filter cartridges; Aquarius's sockets were round. From plastic bags, cardboard, and tape on board, ground engineers devised an adapter, then talked the crew through assembling it. On April 17th the capsule splashed down in the Pacific.

Yet the fault had been built in on Earth. A NASA review board traced the explosion to the tank's thermostat switches, designed for the ship's 28-volt supply. In 1965 the spacecraft's builder rewrote the specification to let the heaters run on 65-volt launch-site power; the switches were never upgraded, a mismatch neither that contractor nor NASA detected. Weeks before launch, to drain the stubborn tank, a heater ran about eight hours. The switches probably malfunctioned; unchecked heat scorched the insulation on the tank's wiring; in flight, that insulation ignited.

NASA called Apollo 13 a "successful failure". The paradox holds: a flight undone by a five-year-old engineering oversight was salvaged within four days by ingenuity on the ground.

中文译文

1970年4月11日，NASA发射阿波罗13号，原计划完成第三次载人登月。4月13日，飞船已飞出约20万英里，服务舱内一个氧气罐爆炸。飞船大部分氧气和电力都由服务舱提供；爆炸使航天器严重受损，登月任务被取消。

宇航员转入“水瓶座”号登月舱避难，而它原本只能维持两个人生存约两天。机组用登月舱发动机调整航向，使飞船沿绕月轨迹飞行，再借月球引力转向地球。他们限量用水、节约电力，在严寒的舱内瑟瑟发抖。

另一种不易察觉的危险出现了：舱内多了第三个人，二氧化碳不断积聚。机组乘坐的指令舱使用方形滤芯，水瓶座号的插槽却是圆的。地面工程师利用飞船上现有的塑料袋、硬纸板和胶带设计转接装置，再通过通话一步步指导机组组装。4月17日，指令舱在太平洋溅落。

然而，隐患早在地球上就已埋下。NASA审查委员会将爆炸原因追溯到罐内的温控开关：它们按飞船28伏电源设计。1965年，航天器承建商修改技术规格，允许加热器使用发射场65伏电源，开关却始终没有升级。承包商和NASA都未发现这种不匹配。发射前几周，为排空那个难以排空的储罐，一台加热器连续运行约八小时。温控开关很可能失灵，失去控制的高温烧焦罐内电线的绝缘层；飞行途中，绝缘材料起火。

NASA把阿波罗13号称为“成功的失败”。这看似矛盾的评价确实成立：一次飞行毁于五年前的工程疏忽，却在四天内靠地面人员的巧思得以挽救。

核心词汇

先记这22项。连着下方的原文短语一起回想。

cripple v.
使严重受损
The blast crippled the spacecraft

abort v.
中止；取消
the landing was aborted

refuge n.
避难；庇护
took refuge in the lunar module

trajectory n.
轨迹；飞行路线
a trajectory around the Moon

ration v.
限量供应；限量使用
They rationed water

hazard n.
危险；隐患
A quieter hazard emerged

emerge v.
出现
A quieter hazard emerged

accumulate v.
积聚
carbon dioxide was accumulating

devise v.
设计出；想出
ground engineers devised an adapter

adapter n.
转接装置
devised an adapter

assemble v.
组装
through assembling it

fault n.
缺陷；故障
the fault had been built in on Earth

thermostat n.
温控器
the tank's thermostat switches

specification n.
技术规格
rewrote the specification

mismatch n.
不匹配
a mismatch

drain v.
排空
to drain the stubborn tank

malfunction v.
失灵；发生故障
The switches probably malfunctioned

insulation n.
绝缘材料
the insulation on the tank's wiring

paradox n.
看似矛盾却成立的事
The paradox holds

oversight n.
疏忽
a five-year-old engineering oversight

salvage v.
挽救
was salvaged within four days

ingenuity n.
巧思；足智多谋
by ingenuity on the ground

文中词汇与搭配

补充词汇共52项。保留每个表达在本文中的意思。

launch v.
发射

crewed adj.
载人的

lunar landing 名词短语
登月

oxygen tank 名词短语
氧气罐

explode v.
爆炸

service module 名词短语
服务舱

blast n.
爆炸；爆炸冲击

spacecraft n.
航天器

astronaut n.
宇航员

take v.
采取；与refuge连用指避难

in prep.
在……中；在此引出避难处

lunar module 名词短语
登月舱

engine n.
发动机

steer v.
操纵；引导航向

craft n.
航天器

onto prep.
进入；在此指进入某条轨迹

swing v.
使转向；甩回

conserve power 动词短语
节约电力

shiver v.
因寒冷而发抖

bitterly adv.
极其；非常

cabin n.
舱室

carbon dioxide 名词短语
二氧化碳

capsule n.
太空舱；返回舱

filter n.
过滤器

cartridge n.
可替换的滤芯

socket n.
插槽；接口

文中词汇与搭配 续

补充词汇共52项。保留每个表达在本文中的意思。

cardboard n.
硬纸板

tape n.
胶带

talk v.
口头指导；与through连用

through prep.
贯穿整个过程；与talk连用

splash down 动词短语
在水上溅落

build in 动词短语
预先嵌入；埋下

review board 名词短语
审查委员会

trace v.
追溯；查明来源

explosion n.
爆炸

to prep.
用于引出追溯到的原因

switch n.
开关

volt n.
伏特；电压单位

builder n.
承建商

rewrite v.
改写；修改

heater n.
加热器

run on 动词短语
靠某种能源运行

upgrade v.
升级

contractor n.
承包商

detect v.
发现；查出

stubborn adj.
难处理的；此处指难排空的

heater n.
加热器

unchecked adj.
未受控制的

scorch v.
烧焦

wiring n.
电线；线路

ignite v.
起火；燃烧

be undone by 动词短语
被……毁掉；因……而失败

回想与回顾

先遮住下方答案，再把四个词放回原文。

01 The blast crippled the spacecraft; the landing was _____.

02 They _____ water, conserved power, and shivered in a bitterly cold cabin.

03 A quieter _____ emerged: with a third man aboard, carbon dioxide was accumulating.

04 a flight undone by a five-year-old engineering _____

文章主线

- 事故 1970年，阿波罗13号氧气罐爆炸，登月取消，三名宇航员转入原本只供两人约两天的登月舱。
- 险情 机组缺水缺电、忍受严寒，二氧化碳不断积聚，清除它的方形滤芯却装不进圆形插槽。
- 解围 地面工程师用船上的塑料袋、纸板和胶带设计转接器，指导机组组装，让滤芯接上圆形插槽。
- 祸根 发射前，加热器用65伏地面电源，温控开关仍按28伏设计；开关疑失灵，约八小时加热损伤线路绝缘层，飞行中起火。
- 返航 机组用登月舱发动机调整绕月路线，借月球引力返航；爆炸四天后，三人安全溅落太平洋。

参考答案　1 aborted 2 rationed 3 hazard 4 oversight